

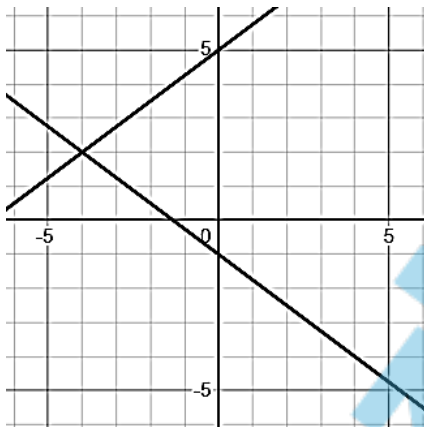
SAT UZBEKISTAN

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December Math

22 QUESTIONS

1



The graph of two linear equations is shown. The solution to the system is (x, y) .

What is the value of y .

- A) -4
- B) -1
- C) 2
- D) 5

2

The function f is defined by $f(x) = 45x + 30$. What is the value of $f(x)$ where $x = 2$?

- A) 77
- B) 90
- C) 120
- D) 150

3

In the xy -plane, the graph of $y = f(x)$ is a line that passes through $(0, 28)$ and has a slope of 5. Which equation defines f ?

- A) $f(x) = 28x$
- B) $f(x) = 28x + 5$
- C) $f(x) = 5x$
- D) $f(x) = 5x + 28$

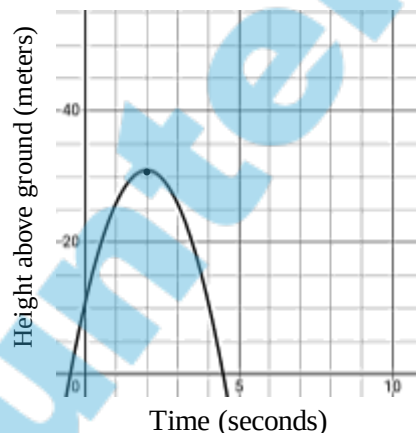
4

x	y
0	23
1	24
2	25

The table shows three values of x and their corresponding values of y . There is a linear relationship between x and y . Which of the following equations represents this relationship.

- A) $y = 23x$
- B) $y = 25x + 2$
- C) $y = 25x$
- D) $y = x + 23$

5



An object was launched upward from a platform. The graph shown models the height above ground, y , in meters, of the object x second after it was launched. For which of the following intervals of time was the height of the object increasing for the entire interval?

- A) From $x = 0$ to $x = 2$
- B) From $x = 0$ to $x = 4$
- C) From $x = 2$ to $x = 3$
- D) From $x = 3$ to $x = 4$

6

The relationship between the variables x and y is defined by an exponential equation. When $x = 0$, the value of y is 40, and for every increase in the value of x by 1, the corresponding value of y increases by 20% of its previous value. Which equation represents this relationship?

- A) $y = 40(1.20)^x$
- B) $y = 40(1.02)^x$
- C) $y = 20(1.40)^x$
- D) $y = 20(1.04)^x$

7

Task	Time (minutes)
A	5
B	4
C	11
D	10
E	10

The table shows the amount of time it took a participant in a study to complete each of 5 tasks. What was the mean time, in minutes, for this participant to complete a task?

- A) 7
- B) 8
- C) 10
- D) 11

8

The function h is defined by $h(x) = 7|x|$. What is the value of $h(-5)$?

- A) -35
- B) 2
- C) 12
- D) 35

9

x	$h(x)$
0	1.46
2	2.32
4	3.68

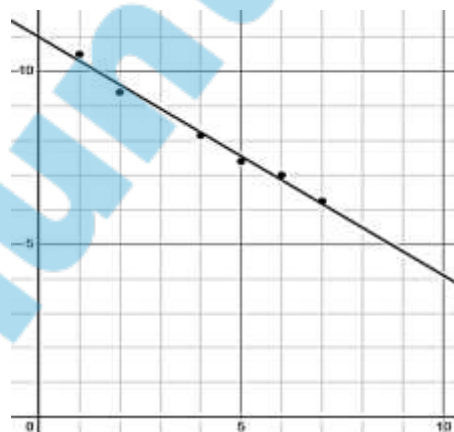
The table shows the exponential relationship between the number of years, x , since Isela started training in pole vault, and the estimated height $h(x)$, in meters, of her best pole vault for that year.

Which of the following functions best represents this relationship, where $x \leq 4$?

- A) $h(x) = 1.26(0.46)^x$
- B) $h(x) = 1.26(1.46)^x$
- C) $h(x) = 1.46(0.26)^x$
- D) $h(x) = 1.46(1.26)^x$

10

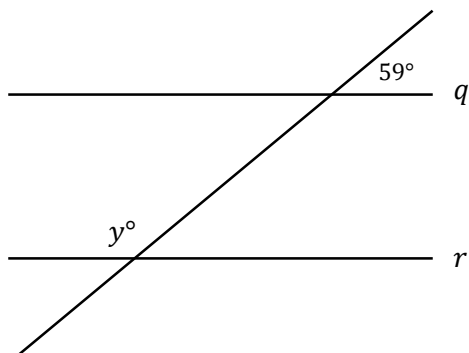
The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.



Which of the following is closest to the slope of the line of best fit shown?

- A) 10.95
- B) 0.69
- C) -0.69
- D) -10.95

11



In the figure, line q is parallel to line r , and both lines are intersected by line s .

If $y = 2x + 9$, what is the value of x ?

- A) 25
- B) 34
- C) 56
- D) 66

12

Line t in the xy -plane has a slope of $-\frac{1}{2}$ and passes through the point $(8, 3)$. Which equation defines line t ?

- A) $y = 7x - \frac{1}{2}$
- B) $y = 8x + 3$
- C) $y = -\frac{x}{2} + 3$
- D) $y = -\frac{x}{2} + 7$

13

$(x - 20)(x - 12)(x + 5)(x + 17) = 0$
What is a positive solution to the given equation?

- A) 20 B) 15 C) 18 D) 30

14

A right circular cone has a height of 12 centimeters (cm) and a base with a radius of 4 cm. What is the volume, in cm^3 , of this cone?

- A) 16π
- B) 48π
- C) 64π
- D) 192π

15

A linear model estimates the population of a city from 1992 to 2016. The model estimates the population was 51 thousand in 1992, 225 thousand in 2012, and x thousand in 2016. To the nearest whole number, what is the value of x ?

- A) 258 B) 259 C) 260 D) 261

16

$$y = 500(3)^{2x}$$

The equation gives the estimated number of online newsletter subscribers at the end of every six-month period, x , starting from the end of January 1992, where $0 \leq x \leq 4$. Which statement is the best interpretation of the y -intercept of the graph of this equation in the xy -plane?

- A) The estimated number of online newsletter subscribers at the end of the first six-month period was 500.
- B) The estimated number of online newsletter subscribers at the end of January 1992 was 500.
- C) The estimated number of online newsletter subscribers at the end of January 1993 was 1,500.
- D) The estimated number of online newsletter subscribers at the end of January 1992 was 1,500.

17

$$\begin{aligned} \frac{x}{3} + 8(y - 16) &= 27 \\ \frac{x}{5} - 8(y - 16) &= 45 \end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $8x$?

- A) 960 B) 1080 C) 1020 D) 980

18

When several resistors are connected in series in a circuit the total resistance of the circuit is the sum of the resistances of these resistors. A certain circuit consists of 6 resistors connected in series, where the resistance of each resistor is positive. This circuit has a total resistance of 80 ohms. The total resistance of 2 resistors in this circuit is 60 ohms. Which inequality best represents all possible values of the resistance x , in ohms, of one of the other 4 resistors in this circuit?

- A) $0 < x < 5$
- B) $0 < x < 20$
- C) $20 < x < 60$
- D) $20 < x < 80$

19

Jasmin grows bean pods in her garden. This year, she harvested 490 bean pods and saved 10% of them to plant next year. How many of the harvested bean pods did Jasmin save to plant next year?

- A) 39
- B) 49
- C) 57
- D) 59

20

A family has money in an account for renting movies online. Each time the family rents a movie, the cost of the rental is withdrawn from the account, and each rental costs the same amount of money. The function $f(m) = 24 - 6m$ gives the amount of money, in dollars, in the account after the family has rented m movies.

Which of the following represents the amount of money, in dollars, withdrawn from the account each time the family rents a movie?

- A) $6m$
- B) 6
- C) 24
- D) $24 - 6m$

21

If $9 - 6(3 - 4x) = 6 - 7(3 - 4x)$,
what is the value of $3 - 4x$?

- A) -3 B) 3 C) -6 D) 6

22

On a plot of land, 52.0% of the square footage is farmland and the remaining square footage is pasture. There are buildings on exactly 21.5% of the square footage of the farmland, and there are buildings on exactly 14.0% of the square footage of the pasture. If there are buildings on exactly $p\%$ of the square footage of the plot of land, what is the value of p ?

- A) 17.6 B) 17.8 C) 17.9 D) 21.6

This module is over.
Do not turn over the page until your time is
up for module 1.

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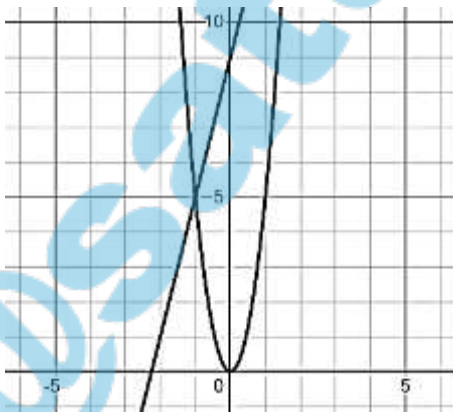
22 QUESTIONS

1

Levi and Marissa each have stamp collections. The number of stamps in Levi's collection is 200% of the number of stamps in Marissa's collection. If there are 196 stamps in Marissa's collection, how many stamps are in Levi's collection?

- A) 98
- B) 196
- C) 392
- D) 588

2



The graph of a linear equation and a quadratic is shown. Which of the following is a solution (x, y) to the system.

- A) $(-1, 5)$
- B) $(0, 0)$
- C) $(0, 9)$
- D) $(1, 5)$

3

An object moves at a speed of $\frac{3}{52}$ feet per second. What is this speed, in yards per second? (3 feet = 1 yard)

- A) $\frac{1}{52}$
- B) $\frac{9}{52}$
- C) 6
- D) 52

4

Which expression is equivalent to $158t^3 - 36t^2u$?

- A) $2t(79t^2 - 18u)$
- B) $2t^2(79t - 18u)$
- C) $2tu(79t^2 - 18)$
- D) $2t^2u(79t - 18)$

5

Circle K has a radius of 11 millimetres (mm). Circle L has an area of $144\pi \text{ mm}^2$. What is the total area, in mm^2 , of circles K and L ?

- A) 23π
- B) 46π
- C) 92π
- D) 265π

6

Which expression is equivalent to $2\left(x - \frac{7}{2}\right)(x + 6)$?

- A) $x^2 + 5x - 21$
- B) $2x^2 - x - 21$
- C) $2x^2 + 5x - 42$
- D) $2x^2 - 42$

7

The function f is defined by $f(x) = (-6)(4)^x + 35$. What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

- A) $(0, 29)$
- B) $(0, 4)$
- C) $(0, 35)$
- D) $(0, -6)$

8

If $4x + 7 = 15$, what is the value of $8x - 3$?

- A) 13 B) 15 C) 18 D) 21

9

$$y < 24 - 8x$$
$$\frac{y}{8} > 9$$

Which inequality represents the x values for all solution (x, y) that satisfy the given system of inequalities in the xy -plane?

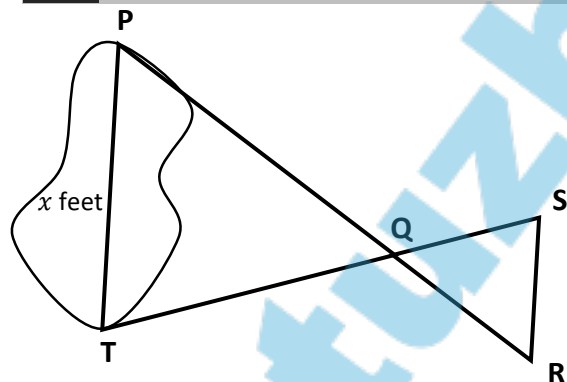
- A) $x > 3$
- B) $x < 3$
- C) $x > -6$
- D) $x < -6$

10

In the xy -plane, an equation of circle A is $(x - 4)^2 + (y - 2)^2 = 9$. Circle B has the same center as circle A but has a radius that is twice the radius of circle A. Which equation represents circle B?

- A) $(x - 4)^2 + (y - 2)^2 = 18$
- B) $(x - 4)^2 + (y - 2)^2 = 36$
- C) $(x - 4)^2 + (y - 2)^2 = 54$
- D) $(x - 4)^2 + (y - 2)^2 = 81$

11

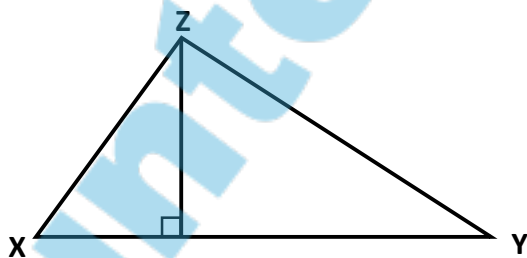


A park designer wants to find the length x , in feet, across a pond as represented in the figure.

PR intersects ST at point Q , and angle PTQ is congruent to angle RSQ . The lengths represented by PQ , TQ , QS , and RS were determined to be 6,300 feet, 10,800 feet, 1,200 feet, and 1,800 feet, respectively. What is the value of x ?

- A) 9600 B) 11200 C) 16200 D) 10800

12



Note: Figure not drawn to scale.

In the figure shown, the measure of angle X is 53° . The length of XY is 24 units and the length of XZ is 18 units. What is the area, in square units, of triangle XYZ ?

- A) $432 \sin 53^\circ$
- B) $216 \sin 53^\circ$
- C) 456
- D) 288

13

$$y = -1.5$$

$$y = x^2 + 4x + a$$

In the given system of equations, a is a positive integer constant. The system has no real solutions. What is the least possible value of a ?

- A) 1 B) 2 C) 3 D) 4

14

A partially filled container containing 26 milliliters of water is placed under a leaky faucet that produces one 0.08-milliliter drop of water every 8 seconds. Until the container is full, which of the following can be used to represent the volume v , in milliliters, of water in the container t seconds after it is placed under the faucet, where t is a multiple of 8?

- A) $v = 0.01t + 26$
- B) $v = 0.08t + 26$
- C) $v = 0.64t + 26$
- D) $v = 8t$

15

A research manager selected 2 random samples of ovens of a certain type to estimate the average amount of time this type of oven takes to preheat to 325 degrees Fahrenheit ($^{\circ}\text{F}$). The research manager recorded the amount of time, in minutes, each oven takes to preheat to 325 $^{\circ}\text{F}$. Based on the first sample, the research manager estimated that this type of oven takes an

average of 14.3 minutes to preheat to 325 $^{\circ}\text{F}$, with an associated margin of error of 1 minute. Based on the second sample, the research manager estimated that this type of oven takes an average of 14.5 minutes to preheat to 325 $^{\circ}\text{F}$, with an associated margin of error of 2.1 minutes. Assuming the margins of error were calculated the same way, which of the following best explains why the first sample obtained smaller margin of error than the second sample?

- A) The first sample contained fewer ovens than the second sample.
- B) The first sample contained more ovens than the second sample.
- C) D) The first sample took less time on average to preheat to 325 $^{\circ}\text{F}$ than the second sample.
- D) D) The first sample took more time on average to preheat to 325 $^{\circ}\text{F}$ than the second sample.

16

$$-\frac{2}{19}rx + \frac{s}{8} = 14 - \frac{7}{57}x$$

In the given equation, r and s are constant. The equation has no solution. What is the value of r ?

- A) $\frac{5}{6}$ B) 1 C) $\frac{7}{6}$ D) $\frac{11}{6}$

17

$$-3x(x + 7) = 27$$

One solution to the given equation can be written as $x = -\frac{s+\sqrt{t}}{2}$, where s and t are positive integers. What is the value of $\frac{s}{t}$?

A) $\frac{7}{85}$

B) $\frac{7}{40}$

C) $\frac{7}{13}$

D) $\frac{14}{13}$

18

The equation $N(m) = 61(Q)^{\frac{m}{4}}$ gives the predicted population $N(m)$, in thousands, of a certain bacteria colony m minutes after the initial measurement, where Q is a constant greater than 1. The predicted population increases by $p\%$ every 120 seconds. What is the value of p in terms of Q ?

A) $100(Q^{\frac{1}{2}} + 1)$

B) $100(Q^{30} + 1)$

C) $100(Q^{\frac{1}{2}} - 1)$

D) $100(Q^{30} - 1)$

19

A quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object was launched. According to the model, the object was launched from a height of 0 feet and reached its maximum height of 1,936 feet 11 seconds after it was launched. Based on the model, what was the height, in feet, of the object 5 seconds after it was launched?

- A) 360 B) 1340 C) 1360 D) 1300

20

The function p is defined by $p(x) = a((x + 5)^2 - b)((x + 5)^2 - c)$, where a , b and c are constants. In the xy -plane, the graph of $y = p(x)$ passes through the points $(-6, 35)$ and $(0, 323)$. What is the value of $p(-10) + p(-4)$?

- A) 323 B) 358 C) 288 D) -14

21

$$19(x - n) = 19y + 19n$$

One of the equations in a system of two linear equations is given where n is a constant. The system has no solution. Which equation could be the second equation in this system?

- A) $2x - 2y = 4n$
B) $2x + 2y = 2n$
C) $2x + 2y = 4n$
D) $2x - 2y = 2n$

22

$$(x - k)^2 = (k - 5a)(x - k)$$

In the given equation, a and k are constants, where $k > 5a$. The sum of the solutions to the equation is $3k + 21$. What is the value of a ?

- A) $21/5$ B) $-21/5$ C) $3/5$ D) $-3/5$